Development of an Optical Tracking System for Real-time Robotic Applications

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Abstract

Engineering students often need a clear, practical model for writing dissertations; this document provides an overview tailored to final-year mechatronics projects and demonstrates effective use of a LaTeX thesis template. Many student reports lack consistent structure, rigorous scoping, and appropriate use of figures, tables, code listings, and citations, which impedes readability and assessment.

This work contributes an exemplar that explains what each report section should contain, offers typical subheadings, and embeds working examples of template elements to copy and adapt. The document is implemented in XeLaTeX using the `memoir` class with chapters for Introduction, Literature Review, Methodology, Results, Discussion, Conclusions, Recommendations, and Appendices, along with acronyms and bibliography support; the result is a shareable guide that doubles as a starting point for students to produce professional, evidence-based engineering reports.

How to write an abstract in 5 minutes?

Answer the following questions in narrative form with no more than two sentences per question:

- *What is the background to the project?*
- *What is the problem that is being solved?*
- *What is the contribution of the project to this problem?*
- *What is the main conclusion of the project?*

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Acronyms

AGP Art Gallery Problem 8–10, 14, 17

BIP Binary Integer Program 10–13

DNA deoxyribonucleic acid 18

FOV Field of View 8

GA Genetic Algorithm 10, 11, 13

GDOP Geometric Dilution of Precision 11, 12, 14

MILP Mixed-Integer Linear Program 10, 11

MoCap Motion Capture 8, 9, 11, 12, 14, 15

NP-hard Non-deterministic Polynomial-time hard 7

OCP Optimal Camera Placement vi, 7–11, 13, 14, 16

PSO Particle Swarm Optimisation 7, 10, 11, 13

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Optimal Camera Placement 5

The quality of a multi-camera localisation system is determined as much by the geometric arrangement as by hardware. Sensor noise, calibration accuracy, and algorithmic performance all matter; yet even a perfectly calibrated, high-quality camera system running a sophisticated algorithm can still fail to localise the target if its cameras are arranged poorly. Determining the Optimal Camera Placement (OCP) requires both a model of how error arises and subsequently propagates, and an effective optimsation formulation that is capable of navigating a large search space timeously.

When the sensor placement is suboptimal, the inaccuracy tends to compound in two ways. The first results from the geometric measurement uncertainty. Cameras discretise their worldview into pixels; the target is projected onto a finite grid of pixels, and its position is inferred from this quantised image (as explained in **??**). The further a target is from a camera, the fewer pixels it subtends, reducing the spatial resolution and increasing the resulting position estimate's variance. Moreover, when two cameras observe a target from nearly parallel viewing angles, small pixel-level uncertainties preclude triangulation [1]. The resulting depth estimate becomes ill-conditioned, degrading the tracking prediction. Hence, in these avoidable configurations, the limitation of the tracking system is not the sensor itself but the geometry.

The second degrading source is visibility loss. Intuitively a camera cannot measure what it cannot see. Occlusion, whether caused by invariant static obstacles in the environment or dynamically occurring by the target itself reduce the number of measurements [2]. Similarly, limited coverage may result in area's being unobserved. Therefore, it may be necessary to optimise for an arrangement that rewards redundant views whilst still ensuring that most of the flight space is observable [3]. Thus, the arrangement of the sensors can exacerbate localisation error beyond the intrinsic noise characteristics of the cameras themselves.

Optimising placement, however, is combinatorially explosive, and classified as Non-deterministic Polynomial-time hard (NP-hard) [4]. As the number of inputs (camera, candidate placements, and workspace size) grow, the number of possible configurations grows exponentially; consequently, an exhaustive search becomes intractable. In this context, NP-hard denotes that the problem becomes increasingly computationally challenging to find a guaranteed global solution with a multitude of inputs. The literature suggests that heuristic- or metaheauristic-based algorithms, such as genetic algorithms, greedy methods, or Particle Swarm Optimisation (PSO) approaches, are adopted because of their proficiency in exploring large, unsmooth,

multi-objective search spaces efficiently [Citation needed]. Alternatives, like integer and mixed-integer programming attempt to find globally optimal solutions but under more heavily imposed constraints. Therefore, each optimisation technique reflects a tradeoff between optimality guarantees, scalability, efficiency and modelling fidelity.

5.1 Literature Review

Optimising camera placement is not optional; it should be a prerequisite for accurate tracking. This section reviews the literature on the OCP problem. First, the optimisation strategies employed to search the ill-defined, non-convex search space in sufficiently good time are examined. Second, an examination of how the sources of measurement uncertainty, quantisation, triangulation geometry, and occlusion, have been characterized and modelled is provided. Together, this provides the state of the art in designing geometrically valid and computationally reasonable multi-camera tracking systems.

5.1.1 Background of OCP

OCP descends from the well-studied Art Gallery Problem (AGP), which established the combinatorial foundations of visibility-based sensor placement. The AGP minimises the number of omnidirectional vertex guards required to observe an entire 2D polygon with n vertices. Fisk proved, via triangulation and 3-colouring, that $\lfloor \frac{n}{3} \rfloor$ guards are always sufficient for covering any simple polygon, as demonstrated in Figure **??**

[Review O'Rourke (1987) Art Gallery Theorems and Algorithms AGP complexity is NP-hard, OCP is also NP-hard]. However, the AGP's usefulness is bounded by its assumptions: where guard are endowed with unlimited sensing range for 360 ° Field of View (FOV), and perfect visibility. Real cameras violate these assumptions; where physical sensors have a limited range of marker detection, resolution that degrades with distance, a finite FOV, and the target may be occluded during observation.

The AGP is bounded by its approach to solely guarantee that all the points in the workspace are seen by idealised observers. Whereas, the OCP objective is to determine the most accurate camera configuration to accurately reconstruct the points. This shift highlights how the OCP frames the problem as one to ensure quality of reconstruction rather than feasibility. In Motion Capture (MoCap), a marker that is only visible by one camera cannot be triangulated; Olague and Mohr emphasise how a marker visible by two (anti)parallel cameras can be triangulated but with large uncertainty in the depth direction; and a marker seen by two or more cameras with sufficient angular separation yields a well-conditioned reconstruction [5]. Therefore, although the OCP inherits the NP-hard structure of the AGP it is necessary for the optimisation to encode that every point of interest is visible by a sufficient number of cameras in a geometrically favourable arrangement rather than described as a binary coverage test.

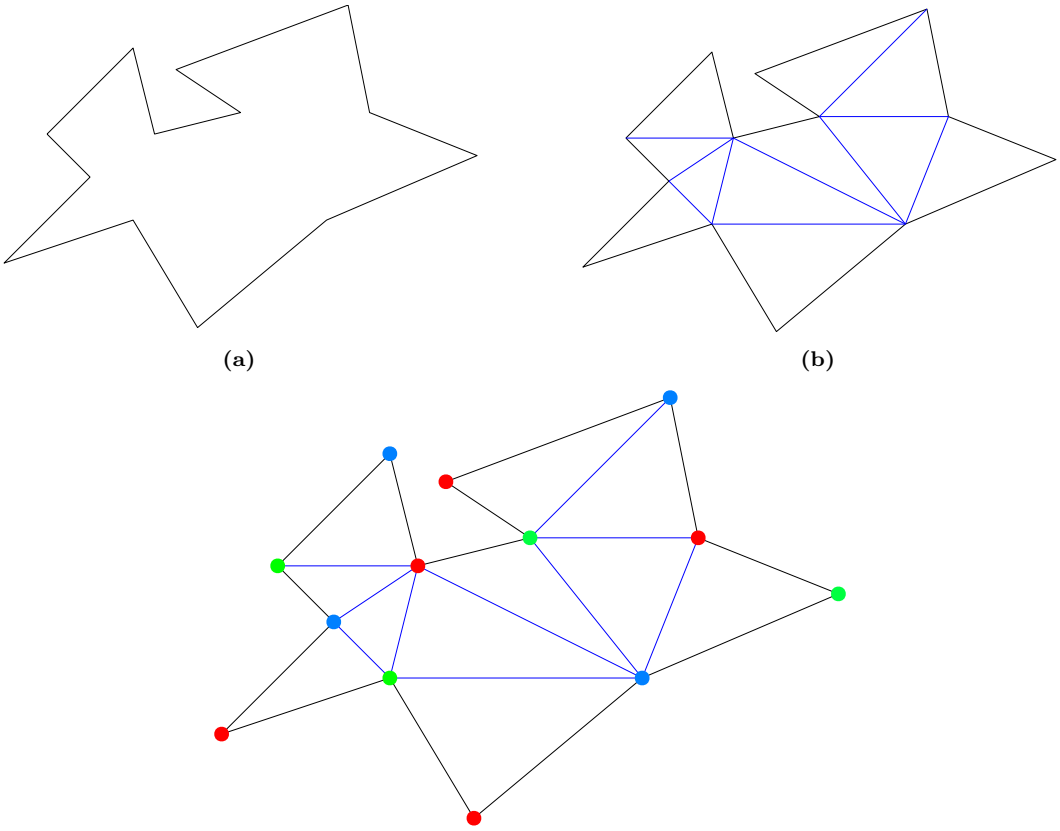

Figure 5.1: In this example, a polygon with $n = 13$ vertices is first triangulated (b), then each vertex is assigned one of three colours such that no triangle shares two vertices of the same colour (c); placing guards at all vertices of the least frequent colour (green or blue) guarantees complete coverage.

5.1.2 Motivation for Optimal Placement

The need for OCP emerges across diverse applications, each imposing different constraints on what embodies optimal placement. Surveillance and monitoring systems are most closely connected to the AGP formulation, where spatial coverage is maximised to ensure that no region goes unobserved, tolerating lower reconstruction accuracy in exchange for exhaustiveness [6]. By contrast, MoCap systems require each marker in the scene to be visible to at least two cameras, with sufficent angular separation simultaneously for accurate triangulation [3, 7, 8]. Autonomous driving scenarios necessitate a full 360° surround view, stitching adjacent camera frustrums to provide the driver and autonomous system with reliable proximity warnings of

neighbouring obstacles [9]. Augmented and virtual reality applications face more severe mounting constraints and an abundance of unpredictable dynamic occlusion, as the human user moves freely within the encapsulating simulation equipment [2]. Despite this diversity, all camera placement optimisers in the literature share the common challenge: the quality of the system's output (surveillance or surrounding map, reconstructed trajectory, or a pose estimate) hinges on where the cameras are placed.

Without a formal optimiser, camera networks are typicaaly set-up through ad-hoc designer intuition or rule-of-thumb guidelines, for e.g. distribute the cameras evenly around the perimeter. While this may appear as an efficient approach, the designer has no means of assessing whether a different arrangement would reduce localisation error. The resulting configuration is neither verifiably optimal nor is the theoretical performance quantified. This is particularly pertinent when the number of cameras is constrained by budget, as each camera should be placed to contribute advantageously to the overall network. Rahimian et al. demonstrated that the performance cost of forgoing optimisation can be substantial. Their PSO OCP algorithm enabled their OptiTrack system to reconstruct all 37 markers on 86% of captured frames, compared to only 30% under an ad-hoc configuration [2]. This nearly tripled improvement achieved by repositioning the cameras in the same room illustrates the performance gap between optimised and unguided placements.

Thus, the motivation to optimise is evident but the problem itself is involved. The AGP, from which the OCP descends, is NP-hard even under its idealised two-dimensional, panoptic-guard assumptions [4]. Subsequently, the OCP is even more complex since it requires more inputs: each camera is described both by a three-dimensional position (x, y, z) and orientation (θ, ϕ, ψ), and the sensor has constraints, such as a finite field of view and depth of field that the AGP does not consider [Citation NEEDED/ section reference here about AGP being NP hard and thus with OCP having more constraints it becomes increasingly challenging]. The placement of k cameras over a subsampled three-dimensional workspace creates a search space that grows combinatorially. If the mounting locations are discretised, further reducing the cardinality of the problem, into N candidate positions, each allowing M candidate orientations, then the number of distinct camera configurations is $\binom{NM}{k}$. For a modest $N = 50$ possible positions, $M = 8$ orientations, and $k = 6$ cameras this exceeds 5^{12} combinations; every one of which requires its own assessment based on the cost function which involves visibility quality and or triangulation geometry computations. Consequently, most investigators resort to meta-heuristic-based approximation algorithms which, albeit without global optimality guarantees, offer the only practical path to obtain acceptable solutions within reasonable computation time.

5.1.3 Optimisation Approaches for Sensor Placement

The OCP literature is conventionally divided into two groups of solver: mathematical programming (typically Binary Integer Program (BIP) or Mixed-Integer Linear Program (MILP)) and metaheuristic search (Genetic Algorithm (GA), PSO, ??).

This framing is not wrong, but it obscures a more consequential distinction. The choice that most constrains both the solver and the fidelity of the resulting solution is whether the placement space is discretised. In a discrete formulation, the continuous room is sampled onto a finite grid of candidate positions and orientations before optimisation begins; the solver then selects a subset of these candidates. In a continuous formulation, the solver operates directly on real-valued camera parameters, position and orientation, without any prior commitment to a grid. Everything downstream, including which algorithms are admissible, what optimality guarantees are available, and how faithfully the cost function can represent the underlying physics, follows from this single modelling decision.

Discrete formulations are the natural domain of mathematical programming. BIP and MILP encode the placement problem as a selection over binary decision variables: each candidate sensor pose is either included or excluded. This structure admits exact solvers such as CPLEX, which exploit branch-and-bound to compute optimality bounds, and permits the designer to prove that a solution is within a known factor of the global optimum [4]. Kirchhof presented both a BIP and a MILP formulation for indoor sensor placement, using Geometric Dilution of Precision (GDOP) as the placement quality metric, and demonstrated that for moderately sized instances the solver can certify optimality. Dai formulated the multi-camera MoCap problem as a BIP with an occlusion culling factor, proving a logarithmic performance ratio $H(k)$ for the greedy sub-solver used to handle larger instances [11]. The cost, however, is modelling fidelity. BIP requires that the objective function, visibility checks, and accuracy metrics all be expressed as linear or quadratic functions of binary variables. In practice, this either limits the sensor model to what can be linearised, or it produces problem instances too large for exact solvers, forcing the use of approximate sub-solvers such as greedy heuristics or LP relaxations even within the mathematical programming framework [**zhaoApproximateTechniquesSolving2013**]. Discretisation also introduces a modelling artefact: the optimal solution is constrained to lie on the grid. The coarser the grid, the larger the potential gap between the discrete optimum and the true continuous one; yet finer grids produce BIP instances that grow combinatorially, which was the motivation for discretising in the first place.

Meta-heuristic algorithms approach the problem from the opposite direction. Rather than requiring the cost function to conform to an algebraic structure, they treat it as a black box: the solver submits a candidate camera configuration and receives a scalar cost, where the optimiser has no knowledge of or requirement on the function's internal form. GAs, PSO, and ?? all operate in this manner, exploring the search space through stochastic, population-based or neighbourhood-based mechanisms that require neither gradient information nor a linearised objective. This formulation-independence is precisely what permits complex cost functions involving ray-cast visibility checks, probabilistic dynamic occlusion models, or GDOP-based triangulation metrics, none of which can be expressed as linear constraints. As shown in Figure **??**, GAs are the most prevalent metaheuristic in the surveyed OCP literature, followed by PSO, with ?? and greedy approaches appearing less frequently. Note that the placement space itself may still be discretised under this paradigm:

Zhou et al., for instance, discretised candidate sensor positions at regular intervals along the workspace perimeter before applying a multi-objective metaheuristic [8]. The distinction between algebraic and black-box solvers is therefore orthogonal to the distinction between discrete and continuous placement spaces, and conflating the two obscures the actual design choices made by each author.

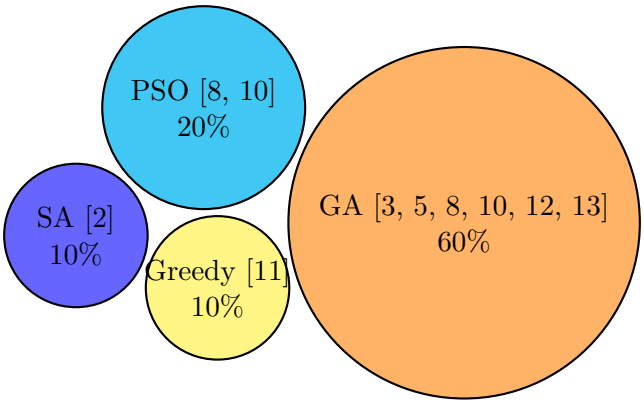

Figure 5.2: Distribution of heuristic-based approaches used for OCP in the reviewed literature

The diversity of solvers in the literature is not arbitrary; each reflects a deliberate trade-off between the guarantees the designer was willing to forgo and the modelling complexity they needed to retain. Kirchhof adopted BIP because indoor localisation with a small number of candidate poses permitted exact optimality certification, and his GDOP metric could be linearised without excessive distortion; but he also presented a continuous NLP alternative for comparison, acknowledging that the discrete formulation imposes a ceiling on solution quality [4]. Dai retained the BIP framework for a larger MoCap problem but conceded the need for a greedy sub-solver once the instance size exceeded what branch-and-bound could handle, proving only a logarithmic approximation ratio rather than exact optimality [11]. At the other extreme, Rahimian and Kearney chose simulated annealing because their occlusion-based error metric, which integrates visibility over all possible orientations of a dynamic occluder whilst simultaneously enforcing triangulability, cannot be decomposed into linear constraints; SA's probabilistic acceptance of worse solutions also provided a mechanism to escape the local minima that pervade such non-convex landscapes [2]. Olague and Mohr selected a real-valued GA for a similar reason: the covariance of 3D point reconstruction that they sought to minimise is a nonlinear function of continuous camera parameters, and the GA's population-based search offered broad exploration without requiring gradient information [5]. Aissaoui et al. adopted a GA but found that an unguided population converged too slowly over the high-dimensional search space of a multi-camera MoCap system; their response was to seed the initial population using a distribution-and-estimation technique that restricted the search to a physically plausible capture zone based on an expected trajectory, producing what they termed a "guided genetic algorithm" [3]. Ziegler et al. avoided meta-heuristics entirely, smoothing the binary visibility function with a tanh approximation to render the problem differentiable and solving the resulting nonlinear program with IPOPT, a gradient-based interior-point solver [7]. The pattern that emerges is instructive: as the cost function retains physical fidelity, the

solver must become more flexible, and the designer must accept weaker guarantees about the quality of the returned solution.

The most revealing trend in the recent literature is not the dominance of any single algorithm but the growing prevalence of hybrid methods. Zhou et al. combined PSO with a GA explicitly to compensate for the tendency of each to stagnate: PSO converges rapidly but is prone to premature convergence in multimodal landscapes, while the GA's crossover and mutation operators maintain diversity but converge slowly on their own [8]. Zhang et al. compared and combined GA and PSO formulations for 3D object coverage [10]. Aissaoui et al.'s guided GA is itself a bit of a hybrid: domain knowledge is used to reduce the effective search space before the stochastic optimiser begins, combining deterministic preprocessing with evolutionary search [3]. That investigators keep bolting supplementary mechanisms onto individual solvers is itself the most compelling evidence that no single algorithm has proven sufficient. The problem resists tidy solutions, and the literature has responded not by finding the right algorithm but by combining imperfect ones.

Returning to the discrete-versus-continuous axis: the literature's treatment of the placement space is seemingly inconsistent. Kirchhof sampled sensor poses at regular intervals along wall edges and mountable surfaces as a prerequisite for his BIP [4]. Dai discretised candidate locations into a grid of positions with associated orientation bins [11]. Zhou et al. placed candidate positions at regular perimeter intervals before applying their hybrid meta-heuristic [8]. In each case, discretisation was adopted to make the problem more computationally manageable, but none of these authors systematically examined what was lost in doing so. On the continuous side, Olague and Mohr encoded camera parameters as real-valued chromosomes [5]; Aissaoui et al. represented each individual as a real-valued vector of translation and orientation parameters [3]; Rahimian and Kearney optimised over continuous camera poses constrained to user-specified geometric regions [2]; and Ziegler et al. solved a continuous nonlinear program directly [7]. Kirchhof is uniquely instructive because he presented both a discrete BIP and a continuous NLP formulation and compared their runtime and solution quality, but this dual treatment remains the exception rather than the norm. What is absent from the literature is a single system that is designed to operate across the spectrum: fully continuous placement when the mounting environment permits it, and discrete placement when physical constraints such as fixed bracket locations demand it, with the same underlying cost function and genetic operators adapted to both modes. (add when the search and mounting space was 3d or constrined to 2d)

5.1.4 Modelling the Problem

maybe restructure the discrete vs contnuous aspect here

5.1.4.1 Optimisation Objectives

The most intuitive OCP objective, and historically the first, is spatial coverage: ensuring that every point in the workspace is observed by at least one camera, or by at least k cameras for redundancy. This objective descends directly from the

AGP, where success is defined as a binary visibility test, a point is either seen or it is not. Bodor et al. maximised an aggregate observability measure weighted by task-specific motion statistics, which is fundamentally a coverage metric, albeit one that prioritises regions of expected activity over uniform spatial exhaustiveness [6]. Zhou etal. included a spatial coverage term in their multi-objective function to ensure that the workspace is adequately monitored [8]. For surveillance applications, where the goal is exhaustive monitoring rather than precise reconstruction, coverage-dominant objectives are often sufficient. For MoCap, they are not. A marker that is visible to only one camera cannot be triangulated. Coverage guarantees observation; it does not guarantee measurement.

For applications requiring three-dimensional reconstruction, the objective must encode measurement quality, and the literature has characterised this quality through three distinct but complementary mechanisms. The first is triangulation geometry. When two cameras observe a target from nearly parallel viewing angles, the resulting depth estimate becomes ill-conditioned regardless of how precisely each camera localises the target in its own image plane. Kirchhof used GDOP explicitly as a metric of geometric quality, penalising configurations where the angular separation between sensor lines-of-sight is insufficient for well-conditioned triangulation [4]. Ziegler et al. constrained the convergence angle between optical axes and demonstrated that triangulation error increases exponentially as the convergence angle approaches 0° or 180° [7]. Olague and Mohr minimised the maximum diagonal element of the covariance matrix of 3D point reconstruction, capturing the worst-case error propagation for a given configuration [5]. The second mechanism is resolution uncertainty, which operates before triangulation at the image-formation stage. Wu, Sharma, and Huang derived analytical uncertainty bounds arising from pixel quantisation as a function of camera-target geometry, showing that the further a target lies from the camera and the more oblique its projection, the larger the variance in the resulting position estimate [1]. The third is dynamic occlusion. Rahimian and Kearney defined a metric that integrates target visibility over all possible orientations of a vertical planar occluder, simultaneously enforcing that the visible camera pairs satisfy a triangulability constraint on their convergence angle [2]. Dai modelled occlusion probabilistically via volumetric occlusion zones with density-dependent visibility, deriving a closed-form approximation for the probability that a marker is seen by at least two cameras [11]. In each case, the designer recognised that the arrangement of cameras,not merely their presence, determines whether the system can produce a usable measurement.

The most recent and arguably most honest formulation of the OCP treats it as an explicitly multi-objective problem, recognising that coverage and accuracy are not independent and that neither alone captures the full cost of a poor configuration. Ziegler et al. maximised the measurement volume whilst enforcing convergence angle constraints, combining a coverage objective with a geometric accuracy requirement in a single optimisation [7]. Zhou et al. constructed a multi-objective function that balanced reconstruction error against spatial coverage, weighted by designer-specified priorities [8]. Aissaoui et al. integrated a composite error metric at the GA evaluation level that combined field-of-view coverage, resolution degradation, and occlusion

into a single weighted fitness score [3]. Several authors further incorporated budget constraints, seeking either to minimise the number of cameras required to achieve a specified quality threshold [4, 11] or to maximise quality for a fixed camera count [2, 3]. The field has converged toward the recognition that single-objective formulations misrepresent the problem. The question is no longer whether to combine objectives but how to weight them, and this weighting inevitably reflects the priorities of the specific application. Table **??** summarises the classification of the reviewed literature across solver type, placement space, and objective category.

Depending on the application, available resources, and workspace constraints, different optimisation objectives have been prioritised in the literature. Broadly, these can be categorised into coverage-based objectives and accuracy-based objectives. In terms of coverage-based objectives, the goal is to maximise the visible area of the workspace, and sometimes going further to handle aspects of occlusion both static and dynamic and hence usually ensuring a two-cover coverage. This is particularly relevant in surveillance applications where ensuring that all regions are monitored is critical [6]. Conversely, accuracy-based objectives focus on minimising localisation error, which is crucial in motion capture and augmented reality applications where precise tracking is required and more than one camera arranged sufficiently non-parallel is needed to ensure triangulation is possible [2, 3]. Wu, Sharma, and Huang proposed another accuracy-based metric of placement which assessed how the limited camera intrinsics and quantisation lead to inaccurate measurements exacerbated by the placement of two cameras [1]. Many studies have proposed hybrid objectives that balance the trade-off of accuracy with coverage [7, 8]. Some works have also considered additional constraints such as budget limits and physical mounting options [3, 9, 11]. The configuration quality for 3D marker tracking and trinagulation is mostly affected by two main sources of error: marker visibility and triangulation accuracy. In terms of the application being MoCap, a marker should not be occluded by any obstacle, including itself, the marker must be within the cameras field of view, vissible given the camera's resolution and range of capture. These are necessary to ensure consistent and continual tracking. Furthermore, at least two cameras should be arranged sufficiently non-parallel so that triangulation is well-conditioned. Sometimes, the easy answer is just to throw more cameras at the problem as that does give redundant views and ensures we can place camera pairs to cover the scene in a sufficently non-parallel format but in most research situationships, it is not prudent to over-exhaust the budget and hence it may be a desirable goal to asses the least amount of cameras required to achieve a level of accuracy/ coverage to minimise cost.

Provided below is a summary table of the different optimisation qualities found across papers in the literature (insert table with rows indication paper citation and application and columns of optimsation objectives)

5.1.5 Summary

The measurement uncertainty attributed to the spatial relationship between the target and the cameras can be modelled based on the distance between the target

Table 5.1: Classification of the reviewed OCP literature by solver, placement space, and optimisation objective.

Source	Solver	Space	Objective	Key Metrics
Kirchhof (2013)	BIP / NLP	Discrete / Continuous	Accuracy	GDOP; min. sensors
Dai	BIP + Greedy	Discrete	Hybrid	Occlusion probability; min. cameras
Bodor et al. (2007)	Hill-climbing	Discrete (refined)	Coverage	Task-weighted observability
Olague & Mohr (2002)	GA (real-valued)	Continuous	Accuracy	Covariance of 3D reconstruction
Aissaoui et al. (2018)	Guided GA	Continuous	Hybrid	Reconstruction error; visibility; resolution; occlusion
Rahimian & Kearney (2017)	SA	Continuous	Accuracy	Dynamic occlusion + triangulability
Zhou et al. (2024)	Hybrid PSO + GA	Discrete	Hybrid	Coverage + reconstruction or

and camera which can excerbated pixel resolution and the orientation between camera's view vectors where the increasingly more parallel these vectors are results in ill-conditioned triangulation geometry, culminating in poor pose estimates and hence poor tracking

As seen there have been a plethora of proposed methodologies for optimising camera placement in the literature which are summarised in the timeline below where they have been broadly categorised as heuristic optimisation or binary integer programming and their objectives have also been listed based on thematic colour rings indicated in the key. .3

The optimal camera placement problem extends the AGP foundation by introducing constraints that reflect the true capabilities and limitations of physical sensors. These extensions transform camera placement from a geometric covering problem into a constrained optimisation problem where the cost function must encode both visibility and measurement quality

Generally, the OCP can be stated as follows: given a set of candidate camera locations and orientations, find the configuration that maximises 3D reconstruction quality of a target within a defined workspace, subject to practical constraints such as mounting options and budget. Solving this problem requires balancing trade-offs among coverage, accuracy, and computational efficiency. Without the aid of an optimisation algorithm, the sensors are arranged in an ad-hoc configuration by the designer, which requires expertise or trial and error, which can be time-consuming or sacrificial if a sub-par arrangement is settled on with no mathematical quantitative guidance. Thus, the resulting localisation error is neither calculated nor considered mathematically. Here lies the need to optimise the camera network based on metrics that minimise degrading effects like resolution uncertainty and probabilistic occlusion even if the trajectory of the target is not known exactly beforehand.

5.2 Theoretical Framework

5.2.1 Genetic Algorithm Terminology

The language used to describe genetic algorithms borrows heavily from evolutionary biology. Table **??** defines these terms in both domains to clarify the analogy and establish consistent vocabulary for the sections that follow.

This biological analogy provides useful intuition: just as natural selection refines a species over generations by favouring advantageous traits, genetic algorithms iteratively improve a population of solutions by favouring those with higher fitness. While genetic algorithms lack the biological complexity of true evolution, they do provide a productive conceptual framework. The effectiveness of this process depends on how solutions are represented, this is often reffered to as the encoding.

Table 5.2: Comparison of biological and genetic algorithm terminology.

Term	Biological Meaning	Genetic Algorithm Meaning
Chromosome	A structure of DNA carrying genetic information	A complete candidate solution encoded in some representation, e.g bit strings
Gene	A unit of heredity determining a specific trait	A single element or parameter within the candidate solution, e.g a bit
Population	A group of interbreeding individuals of the same species	The set of candidate solutions existing at a given iteration
Fitness	An organism's ability to survive and reproduce in its environment	An objective function quantifying how well a candidate solution achieves the desired goal
Generation	All individuals born and living at approximately the same time	One complete iteration of the algorithm producing a new population
Selection	Differential survival and reproduction due to heritable trait variation	An operator that chooses higher-performing individuals to serve as parents
Crossover	Exchange of genetic material between homologous chromosomes during meiosis	An operator that combines portions of two parent solutions to create offspring
Mutation	A random, heritable change in genetic sequence, ususally caused by a erroneous DNA replication	An operator that randomly perturbs one or more genes in a solution

5.2.2 Encoding Schemes

"coming up with the best encoding is tantamount to solving the problem itself" difference between binary encoding, gray code, real encoding, variable length chromosomes

Binary encoding advantages: mutation and crossover are developed based on this type of encoding. limitations: Hamming cliff effect, encoding and decoding, computer based level of precision (binary precision)

Gray code: advantages: removes hamming cliff artefact limitations: not popularised

Real encoding advantages: no encoding and subsequent decoding step required, avoids hamming cliff artefact, more suited for a continous exploration limitations: mutation and crossover need to be restructured explain that this was chosen for implementation -> how mutation and crossover were altered

5.2.3 Selection Methods

Fitness proportionate selection: roulette-wheel selection, stochastic universal sampling number of times an individual is expected to reproduce is equal to its fitness divided by the average fitness in the population roulette wheel: each individual is assigned a slice of the circular roulette wheel

rank-based selection tournament selection boltzmann selection elitism

explain why tournament and elitism were chosen -> monotomic convergence (always improving or consistent best fitness)

5.2.4 Crossover Techniques

single-point crossover double point crossover parameterized uniform crossover

5.2.5 Convergence Analysis

iteration based fitness score based

5.3 Methodology

5.3.1 Algorithm Configuration

5.3.1.1 Parameter Selection

5.3.1.2 Initial Population

5.3.1.3 Tournament Selection

5.3.1.4 Double Point Crossover

5.3.1.5 Mutation

5.3.1.6 Elitism

5.3.2 Cost Function

5.3.2.1 Resolution Uncertainty Cost Function

5.3.2.2 Dynamic Occlusion Cost Function

5.3.2.3 Combined Cost Function

5.3.3 Termination Criteria

5.4 Results and Discussion

5.4.1 Convergence Behaviour

5.4.2 Cost Function Evaluation

5.4.3 Parameter Sensitivity Analysis

5.4.4 Optimal Placement Configurations

Different Modalities Random Placement, ad hoc placement

5.4.5 System Limitations

? future work: if trajectory is known beforehand optimise the placement to be able to best recover the features/ tags by creating clusters of where the expected markers shall be like [13].

Conclusion and Future Work 6

6.1 Summary of Contributions

6.2 Key Findings

6.3 Limitations

includes scope constraints outlined in intro

6.4 Recommendations for Future Work

closed-loop robotic control Benchmark GA against other metaheuristic like PSO or SA to see if better solutions existan assess convergence performance

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